

Independent Summative Evaluation of Competence-Based Curriculum in Kenya's Primary Schools

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In 2017, the government of Kenya adopted the competence-based curriculum approach. Kenya Institute of Curriculum Development conceptualised and developed contextualised or localised competence-based curriculum for primary and secondary schools. The primary school curriculum was implemented nationally in grades 1, 2, 3 in 2019, in grade 4 in 2020, grade 5 in 2021 and grade 6 in 2022. The purpose of this study was to determine the effect of CBC. The study was guided by the following evaluation questions. 1.To what extent was CBC for primary schools systematically developed and implemented? 2.To what extent is the goal of CBC for primary schools appropriate, achievable and realistic? 3.What is the relevance of CBC? 4.To what extent was CBC effectively implemented in primary schools? 5.What are the effects of CBC on parents, pupils and teachers? and 6.What strategies should be employed to strengthen the effectiveness of CBC? A causal comparative evaluation design was used. Data were collected using questionnaires and analysed using frequencies, percentages, mean scores and standard deviations. The following were the main findings of the study: a) CBC was introduced without following the systematic process of curriculum development. b) The CBC for primary schools that was developed for Kenya Institute of Curriculum Development using a localised hybrid CBC approach is appropriate, achievable and realistic. c)Subject areas of CBC are relevant. d)CBC was not effectively implemented. It was hurriedly developed and implemented without being pilot-tested in grades 4, 5 and 6. Teachers have not been sufficiently trained in CBC content, teaching methods and assessment techniques and schools lack adequate equipment, tools and materials for practical skills subjects. e)CBC makes heavy demands on parents and teachers. However, due to implementation of CBC pupils have been equipped with practical skills. On the basis of the findings of this study, the following conclusions are made: i.Adoption of a contextualised CBC approach will improve the quality of education in Kenya. ii.The inadequate training of teachers has hindered many of them from effectively implementing CBC. On the basis of the findings and conclusions of this study, the following recommendations are made: The Ministry of Education should work closely with teacher training institutions in conducting in-service teacher training workshops. The Ministry of Education should provide equipment, tools and materials to schools.

Keywords: Independent summative evaluation, implementation, competence-based curriculum, primary schools

Introduction

Kenya is a lower middle country that is not heavily industrialised. Education is a fundamental element in economic and social development. This is because development requires an educated population. Not only does education provide young people with skills and knowledge they need to achieve a higher standard of living, but also gives them better opportunities for fulfilling themselves as human beings. The following is a summary of the goals of education in Kenya.

National Goals of Education in Kenya

1. National unity education in Kenya must foster a sense of nation hood and promote national unity.
2. National development.

3. Individual development and self-fulfilment.
4. Social equality.
5. Respect and development of cultural heritage.
6. International consciousness.

Since the achievement of independence in 1963, the government of Kenya has been committed to providing relevant quality education and training at all levels and has addressed challenges facing the education sector through education reforms. The first education reform sought to change the education system inherited from the colonial government to make it more responsive to the needs of independent Kenya. (Republic of Kenya, 2005, p.2). Racial schools were abolished and the curriculum was made more relevant to the needs of learners and society.

The second education reform was initiated in 1976 to implement recommendations of the National Committee on Educational Objectives and Policies. The school curriculum was revised to make it terminal. The third major reform was the change from the 7.4.2.3 system of education to the 8.4.4 system of education in 1985. Recent years have seen a plethora of policy initiatives in the education sector in Kenya such as free primary education, a hundred percent transition from primary to secondary school, competence-based curriculum and adoption of 6.3.3 system of education.

The new structure has 2 years of pre-primary education, 6 years of primary education, 3 years of middle school, 3 years of secondary education and a minimum of 3 years of university education.

Overview of the Competence-Based Education and Training Model

Competence is the ability to apply high level knowledge and skills for successful performance of a given task or function. High level knowledge competencies and skills are considered in most countries to be the very basic conditions for citizenship, employment and social cohesion. Based on this definition competence-based curriculum (CBC) is a curriculum approach that develops cognitive abilities, practical skills, attitudes and values.

The competence-based curriculum framework has been embraced by governments around the world. Currently, there are two main competence-based curriculum (CBC) approaches. These are the old CBC approach and the new CBC approach.

The Old CBC Approach. The old CBC is defined as a curriculum that focuses on what learners can do and not on high level cognitive abilities. The main objective of this competence-based curriculum framework is to equip students with different educational attainment to acquire agreed standards of skills appropriate to jobs available and provide them a basis for progression through further learning. Competence-based education and training is seen by many in political and educational circles as a tool to create a workforce capable of underpinning the growing economy. (Mutarabukwa, 2007).

Spady (1978) and other advocates of performance-based curriculum point out that competency involves more than “capacities” such as the ability to read and calculate; it should refer to application of school learned skills in situations outside of school. (Ornstein, p.13). Competence-based training started in the United States of America as vocational training model. The development of competence-based education at training curriculum in the United Kingdom can be traced to a 1981 document called a New Training Initiative. The document specified two goals for reorganisation of training in the United Kingdom. These are:

1. Offering work-based training to all young people aged 18 years and above.

2. Widespread training in different trades to enable vocational training institutions graduates to have wide chances of gaining employment. (Franklin 1997, Mutarabukwa 2007).

The Development. This model is appropriate for technical training whose primary objective is the development of productive skills leading to competent performance of tasks. In technical and vocational training institutions production, knowledge and skills are imparted by individuals with adequate training and interest in the subjects they teach. However, in primary and secondary schools, practical skills subjects are typically taught by teachers who have no training or interest in the subjects. As a result, these subjects are not taught effectively.

According to Brandt and Tyler (2011, p.13),

The concept is especially useful in vocational education where a particular competency can be broken down through task analysis into its component skills.

In recent years, international organisations such as the World Bank, International Monetary Fund and United Nations International Children's Emergency Fund (UNICEF) have advocated the adoption of this approach and revival of practical subjects as a strategy for fostering rural development and addressing the current problem of youth unemployment.

The New Competence-Based Curriculum Framework. The new competence-based curriculum framework is a contextualised (localised) competence-based framework that simultaneously develops high level knowledge and skills and attitudes. Contextualised hybrid competence-based curricula in most countries have much in common: subject areas include English, national language, mathematics, science, ICT.

Key competencies of the European Union (EU) were born in the 1990s. Based on research and recommendations by the European parliament, competencies were defined as a combination of knowledge, skills and attitudes appropriate for personal fulfilment and development, active citizenship, social inclusion and employment. Available evidence shows that governments in Africa have adopted what has been described as subject specific competencies model of competence-based curriculum. Instead of focusing on practice-oriented goals as recommended by international organisations; they have simultaneously focused on high level knowledge and practice oriented goals.

Competence-Based Curriculum in Kenya

The competence-based curriculum approach was adopted by Kenya following the decision by the East African Community to harmonise their curricula. East African Community countries regarded harmonisation of the curriculum and system of education as one way of promoting movement of services across its five members (EAC, 2007). In 2012 based on

advice from UNICEF, EAC developed a framework requiring member countries to adopt CBC.

In 2017, Kenya Institute of Curriculum Development (KICD) adopted competence-based curriculum added curriculum or subject specific competencies model of competence-based curriculum. The goal of the curriculum is to develop high level cognitive abilities and practical skills. Unlike the previous curriculum which mainly consisted of academic subjects, CBC for primary schools has more than ten subject areas. It is a hybrid curriculum approach that simultaneously develops high level cognitive abilities and teaching of practical subjects such as agriculture, home science, art and craft, health, nutrition and pre-vocational skills. These subjects were also emphasised by the original school curriculum for the 8.4.4 system of education.

The Need for Independent Summative Evaluation of CBC

Few people in Kenya would deny that there is no consensus about CBC. Proponents argue that CBC will benefit pupils because they will get competencies and insist that CBC is here to stay. Opponents insist that CBC is contentious. They complain that teachers have not been adequately trained to teach the new curriculum and implementing CBC is too expensive for poor parents. They argue that the government only pays lip service to fair access for children from poor backgrounds.

Currently, knowledge about CBC effectiveness is not only narrow, due to the failure by the Ministry of Education to conduct summative evaluation, but is also shallow due to heavy reliance on statements by senior government officials. The Ministry of Education did not make efforts to collect and share data on international best practice about CBC implementation issues such as teacher training, professional development of teachers, quality assurance, assessment and evaluation, provision of resources to schools and appropriate level of placement of grades 7, 8 and 9 at a level of the education system and the role of parents in their children's learning.

This academic year grade 6 pupils will complete primary education. A natural step would be for the Kenya Institute of Curriculum Development (KICD) to commission external summative evaluation of competence-based curriculum and present a report to the academic board of KICD. A report of summative evaluation of the pilot phase of competence-based curriculum in Kenya which was sponsored by Kenya Union of Teachers and conducted by a team of senior education experts in Kenya and Tanzania whose findings and recommendations were ignored by senior Ministry of Education would have helped the government to improve CBC.

Instead of implementing CBC as advised by UNICEF, the government should have reviewed the school curriculum for the primary cycle of the 8.4.4 system of education to fully align it with the sessional paper no.12 of 2008. Opponents of CBC favour abolishing CBC altogether. That puts reform in a dif-

ficult position. However, reverting to the old curriculum and school system without conducting summative evaluation may have its limitations. Therefore, evaluation is needed.

It has been a matter of concern to many educationists that there has been a tendency for the Ministry of Education to insist on implementing education policies before arriving at consensus with key stakeholders. There is no evidence of improved education attributable directly to placement of grades for pre-adolescents in secondary schools and implementation of competence-based curriculum; and there is some evidence of improved education attributable directly to placement of grades 7, 8 and 9 in primary and middle schools and the curriculum for the primary school level of the 8.4.4 system of education.

Many Ministry of Education including Kenya Institute of Curriculum Development officials have highlighted positive effects of CBC. However, most of them have not provided evidence from research studies. Yet reports from parents and teachers strongly ..., the need for a comprehensive and rigorous external summative evaluation of competence-based curriculum. This study addressed that need. It examined the effects of CBC on pupils, teachers and parents.

Purposes of the Study

This evaluation exercise has two main purposes. First, in 2017 the government changed the country's system of education from 8.4.4 to 6.3.3.3. This made grades 7 and 8 part of the secondary level. This study investigated the need for this change. In particular, this study will attempt to remedy the lack of a pre 2.6.3.3.3 education structure data.

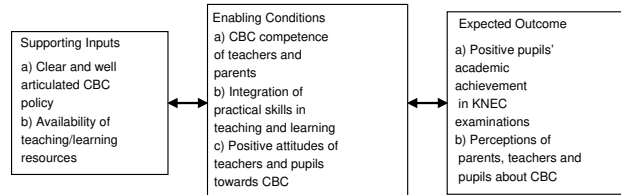
Second, first group of students who were taught CBC are currently in their final year of primary education. There is need to conduct summative evaluation of CBC to provide data to policy makers to help them decide to continue, revise or terminate CBC.

Evaluation Questions

The study will be guided by the following evaluation questions:

1. To what extent was competence-based curriculum for primary schools systematically developed and implemented?
2. To what extent is the goal of competence-based curriculum for primary schools appropriate achievable and realistic?
3. What is the relevance of subject areas of CBC?
4. To what extent was competence-based curriculum effectively implemented in primary schools?
5. What are the effects of competence-based curriculum on children and their parents?
6. What strategies should be employed to strengthen the effectiveness of CBC?

Figure 1. Conceptual Framework for Implementation of CBC to have an Effect on Pupil Academic Achievement



Significance of the Study

It is expected that the findings of this study will assist the government to develop and implement an appropriate curriculum for primary schools. It is also hoped that the study will offer a thorough view of the needs and make recommendations for effective and efficient utilisation of human and financial resources for the purpose of providing quality education.

Conceptual Framework

The evaluators tried to give a prediction on the variables that show the effects of CBC on pupils, teachers and pupils' academic achievement. The effect of CBC is based on the relationship of three variables and positive academic achievements of pupils. The three variables are supporting inputs: Clear and well articulated primary school CBC policy and availability of teaching and learning resources. Enabling conditions: teachers' and parents competence, integration of practical skills in teaching and learning of different subjects and positive attitudes of teachers and pupils towards CBC and expected outcomes: positive academic achievement of pupils, level of teacher's preparedness to implement CBC.

Evaluation Model

This evaluation adopted the context-input-process and product (CIPP) evaluation model which was developed by Stufflebeam (1974). CIPP is a management oriented evaluation model. Management evaluation models lay emphasis on the use of evaluation to facilitate good decision making. The main reason for adopting this model was that the study would provide relevant information to the Kenya Government for evaluating alternative decisions and necessary changes in future development of the school curriculum.

Scope and Delimitations of the Study

The study is essentially an evaluation of the implementation of competence-based curriculum in grades 1 to 6. It does not include evaluation of competence-based curriculum for secondary schools.

Limitations

Finally, the present study was conducted in four counties that were not randomly selected. Respondents from other counties may respond differently from those who participated in this study. Generalisability is limited to schools, parents, head teachers, teachers and pupils represented by the participants in this study. However, generalisability may extend to counties with populations and environments that are similar to those in this study.

Evaluation Design and Methodology

Evaluation Design

This study used *ex-post facto* design. An *ex-post facto* (causal comparative) evaluation design was used in the quantitative study. The ideal designs for summative evaluation are experimental and quasi-experimental designs. In these designs treatment and control group data are collected. Programmes often enlist professional evaluators to conduct summative evaluation.

Rankings based on perceptions are problematic. Respondents may be so used to the wrong things they do that they do not see them as a problem. Therefore the rankings are only used as broad guides. Teachers are normally judged on their pupils' progress as well as evaluations by head teachers and peers.

Target Population

The target population in this study included all parents with children in primary school, head teachers, teachers and community leaders in four counties.

Description of the Sample and Sampling Procedures

The study was funded by members of the evaluation team. Owing to budgetary constraints and time, the study only covered a randomly selected sample of primary schools in four counties, namely, Busia, Uasin Gishu, Turkana and Kiambu.

Research Instruments

Six kinds of instruments were administered to the five types of respondents to determine the effects of CBC. These instruments were the following:

1. Questionnaire/interview schedule for parents
2. Questionnaire for pupils
3. Questionnaire for head teachers
4. Questionnaire for teachers
5. Questionnaire for educators and community leaders
6. Document analysis guide
7. Lesson observation guide

Validity of Research Instruments Results

The validity of research instruments results was determined in terms of content validity. A panel of three experts in education and research analysed both individual items in questionnaires and questionnaires according to the following aspects: relevance and appropriateness of the items to research questions, appropriateness of items for respondents, coverage of domain of interest, clarity of the language used.

Pilot-Testing of Evaluation Instruments

To improve the validity of evaluation instruments results, the instruments were pilot-tested on small sample of parents, head teachers, teachers, pupils, administrators, educators and community leaders. Instruments were revised after the piloting.

Reliability of Research Instruments Results

Reliability of research instruments results was estimated using split-half and test re-test techniques.

Description of Data Collection Procedures

After permission had been obtained from the National Council for Science and Technology, evaluators visited schools in the sample and collected data.

Description of Data Analysis Procedures

Data were analysed using descriptive statistics, frequencies, percentages and mean scores and standard deviations.

Ethical Consideration

Ethical consideration was carefully adhered to by the evaluators. Informed consent, anonymity, confidentiality and honesty were ensured all the time during the study.

Results

The following are the findings of the study.

Extent to which Competence-Based Curriculum for Primary Schools was Systematically Developed and Implemented. CBC was introduced in primary schools without public participation, a constitutional requirement. No education commission was established to collect public views on the need for education reform, no sessional paper or CBC had been submitted to parliament for approval, pilot-testing of the curriculum in grades 1, 2 and 3 was hurriedly done, there was no pilot-testing of curriculum in grades 4 and 6 and no internal summative evaluation of CBC was conducted by Kenya Institute of Curriculum Development. In addition, the Kenya Institute of Curriculum Development constructed tests and asked head teachers to download them. Formative assessment tests are supposed to be constructed by classroom teachers themselves.

Extent to which the Goal of CBC is Appropriate, Achievable and Realistic. As a result of serious research, educators in Europe developed a type of competence-based curriculum that simultaneously focuses on high level knowledge and practical skills. The key competencies are:

Examination of CBC for Kenya's primary schools shows that it consists of the following subject areas. The key competencies are:

This shows that the Kenya Institute of Curriculum Development conceptualized and contextualised the subject specific competencies approach similar to the one currently used in European countries.

Conclusion. The intention of adopting the European of competence-based curriculum by Kenya Institute of Curriculum Development (KICD) was good. However, in contrast to schools in European countries, primary schools in many parts of the country lack adequate teachers, equipment and tools and materials. In addition, teachers have not been adequately trained to effectively implement competence-based curriculum and supervision.

Extent to which the Objectives of CBC have been Achieved. Data analysis in this study has shown that CBC has not achieved its overall objective of developing pupils' competencies or skills. However specific objectives of subject areas have to a large extent been achieved. Despite numerous complaints of lack of adequate in-service training of teachers to enable them to effectively implement CBC and inadequate instructional materials efforts have been made to achieve the objectives of subject areas.

Relevance of Subject Areas of CBC. This section considers the relevance of the subjects/learning areas and content of CBC which pupils covered at the primary school level. Table 1 and 2 present teachers' rating of relevance of subjects of CBC.

The responses in Table 1 show that a vast majority of teachers in the sample regarded all subjects taught in lower primary very relevant and relevant.

Table 2 shows that nearly all teachers considered Kiswahili, Mathematics, Science and Technology and Social Studies very relevant. Practical skills subjects, namely Home Science, Agriculture, Art and Craft and Music were considered very relevant by more than half of the teachers, while foreign languages were rated low.

The responses in Table 3 shows that a vast majority of teachers considered the content of English, Kiswahili, Science and Technology, Mathematics, Religious Education, Social Studies and Physical and Health Education very relevant and relevant whilst the content of indigenous languages was rated lower.

Extent to which CBC has been Effectively Implemented in Primary Schools. Educational research and evaluation studies have consistently shown that low learning outcomes in Kenya and other African countries are not so much the result of the planned curriculum as lack of effective implementation of the curriculum.

The evidence from different sources indicates that:

Table 1
Frequency Distribution of Teachers Rating of Relevance of Subject Areas of CBC

Rating				
Subject Areas	Very Relevant	Relevant	Somewhat Relevant	Not Relevant
1. Kiswahili Language activities	38	3	1	1
2. English Language activities	32	6	3	-
3. Mathematics activities	36	4	2	3
4. Religious Education activities	30	6	3	2
5. Hygiene and Nutrition activities	26	12	2	2
6. Environmental activities	22	14	4	1

1. Parental participation in their children's learning as proposed by CBC is too demanding and costly.

2. Teachers were not given adequate training on the subject matter of practical skills subjects.

3. Most people have not understood the importance of CBC.

4. Schools lack materials, equipment, tools and digital devices to effectively implement CBC because most parents are poor.

5. CBC makes heavy demands on learners and parents.

6. It is not realistic to expect parents to teach their children what they are not familiar with. This may lead to parents misleading their own children. Moreover many parents are semi-literate and cannot participate in their children's learning.

7. CBC does not put enough emphasis on the development of high level cognitive skills as is done in developed countries.

8. There is evidence that CBC is developing learners' ability to use ICT and digital devices and developing pupils' skills in the subjects they learn better than the previous curriculum.

9. Schools lack adequate equipment, tools and teaching/learning materials.

10. Some parents lack confidence in CBC.

11. Teachers' inability to implement CBC due to lack of adequate training.

12. Inadequate infrastructure and facilities.

13. The curriculum emphasises development of learners' competencies but teachers have not been adequately trained to develop practical skills and schools lack adequate equipment,

Table 2
Frequency Distribution of Teachers Rating of Relevance of Subject Areas of CBC: Grade 4-6

Rating				
Subject Areas	Very Relevant	Relevant	Somewhat Relevant	Not Relevant
1. Kiswahili	38	3	1	-
2. English	30	5	4	3
3. Mathematics	34	4	3	1
4. Science & technology	38	2	1	-
5. Religious Education	28	8	4	2
6. Social Studies	35	4	2	1
7. Home Science	26	5	7	4
8. Art & Craft	24	6	6	6
9. Music	24	8	4	6
10. Physical & Health Education	28	4	6	4
11. Indigenous Languages	20	10	8	4
12. Pastoral Programme of Instruction	20	14	6	2

tools and materials for practical skills. Hence CBC has not been effectively implemented. There are a number of reasons why schools lack adequate resources. Principal among these ...

Respondents' Views on Whether CBC should be Retained as it is, Improved and Retained or Terminated.

Parents were next asked about whether CBC should be retained as it is, improved and retained or terminated. Their responses are presented in Table 4.

Head teachers were asked to give their views on whether CBC should be retained as it is, improved and retained or terminated. Below are their responses.

Retained as is 2

Improved and retained 4

They gave the following reasons for their responses.

Retained as it is. It is good.

Improved and retained. Most aspects of CBC are generally good.

Terminated. The curriculum designs are not well organised in terms of scope, work load for different grades and does not consider the mental age of learners. It is not organised from simple to complex.

Effect of CBC on Children. The effect of CBC was supposed to be assessed by administering an achievement test to

Table 3

Frequency Distribution of Teachers Rating of Relevance of Content of CBC Subjects: Grade 4-6

Rating				
Subject Areas	Very Relevant	Relevant	Somewhat Relevant	Not Relevant
1. English	38	2	1	1
2. Kiswahili	40	2		
3. Home Science	24	6	8	4
4. Agriculture	18	10	0	5
5. Science and Technology	39	2	1	-
6. Mathematics	36	4	2	-
7. Religious Education	34	3	3	2
8. Social Studies	30	7	3	2
9. Creative Arts	28	6	5	3
10. Physical and Health Education	32	8	2	-
11. Music	26	10	2	4
Indigenous Languages	16	7	10	7
Pastoral Programme of Instruction	22	10	6	4

pupils in pilot schools and some selected non pilot schools. The achievement was to be based on the work which had been covered in grade 1 to 6. Because CBC was implemented in all schools, the evaluation team decided to use standard 7 pupils who had followed the old curriculum as a comparison group.

The issue to explore was whether CBC is more effective in developing pupils' ability to perform tasks in the subjects they learn than the previous curriculum. Since it was possible to administer practical tests to pupils in CBC and comparison groups, evaluators decided to use pupils' perceptions of their ability to perform tasks in their subjects. The tasks included numeracy skills, English Language, Kiswahili and mother tongue (communication skills, writing, reading, listening and speaking), use of computer and practising skills in the subjects they learn. Table 5 presents mean pupils' perceptions of their ability to perform tasks in the subjects they learn.

Table 5 shows that pupils studying CBC had a higher mean perception score (38.4) than those studying the previous curriculum. This is mainly because some of the use computers and other digital devices. The mean score for CBC was very different from the mean score for non CBC schools. The difference was statistically significant.

Strategies that should be Employed to Strengthen the Effectiveness of CBC. Finally, respondents were asked to

Table 4

Respondents Views on Whether CBC should be Retained as it is, Improved and Retained or Terminated

	Retained as is	Improved and Re-tained	Terminated	Total
1. Parents	6	24	16	46
2. Head teachers	2	4	2	8
3. Administrators, community leaders and professionals	0	14	2	16
4. Teachers	2	33	7	42
5. Pupils studying the previous curriculum	2	6	44	52

Table 5

Mean Pupils' Perceptions of their Ability to Perform Tasks in the Subjects they Learn (Out of a maximum score of 50).

	n	Mean	Standard Deviation
Pupils studying CBC	56	38.4	3.25
Pupils studying the previous curriculum	52	24.6	4

suggest strategies that should be used to strengthen the effectiveness of CBC. This section summaries respondents' suggestions on strategies for bringing about effective implementation of CBC in primary schools. Respondents cited a number of strategies that should be employed to strengthen the effectiveness of CBC. Put together, nine main strategies emerged. These are listed as follows:

- Changing the focus of the school curriculum from skills to development of higher intellectual skills such as reasoning, analysis, evaluation, problem solving and critical thinking.
- Develop a comprehensive programme of in-service training of teachers.
- Reduce parental involvement in their children's learning.
- The government should supply materials, tools and equipment.
- Employ more teachers.
- Teachers should establish respectful and productive relationships with parents and guardians.
- Schools should have adequate teachers when revising the school curriculum.
- Curriculum developers should be responsive to children's family circumstances and the needs of children in socioeconomically disadvantaged areas of the country.
- Parents should be sensitised to the importance of involvement in their children's education.

Summary, Conclusions and Recommendations

Summary

The purpose of the study was to determine the effect of CBC. The study was guided by the following evaluation questions:

1. To what extent was competence-based curriculum for primary schools systematically developed and implemented?
2. To what extent is the goal of competence-based curriculum for primary schools realistic, appropriate, achievable and realistic?
3. What is the relevance of content of CBC?
4. To what extent was competency-based curriculum effectively implemented in primary schools?
5. What are the effects of CBC on parents, pupils and teachers?
6. What strategies should be employed to strengthen the effectiveness of CBC?

Evaluation adopted Daniel Stufflebeam's CIPP evaluation model. A causal comparative evaluation design was used. Due to shortage of funds and time, the study was conducted in four counties only; Busia, Turkana, Uasin Gishu and Kiambu. The sample consisted of 42 parents, 8 head teachers of primary schools, 42 teachers, 56 grade 6 pupils and 52 grade 7 pupils and community leaders, administrators and professionals. Data were collected by means of questionnaires. Returns were received from the four counties which provide the information for this report. Data were analysed using frequencies, percentages and mean score.

Main Evaluation Findings

Extent to which CBC was systematically developed and implemented. Using a systematic process of curriculum development:

1. Competence-based curriculum was introduced in Kenya without public participation and was hurriedly implemented in grades 4, 5 and 6 without pilot-testing curriculum materials in few schools. Pilot-testing of CBC in grades 1, 2 and 3 was hurriedly done. Curriculum reform should be based on a detailed diagnosis of past trends, the status, functioning and results of the existing curriculum designed to identify its strengths, weaknesses and opportunities for improvement. It also includes a critical examination of international best practices. It is unfortunate that CBC was introduced without going through this stage. Introducing a new curriculum without establishing an education commission to get the views of stakeholders has a potentially damaging effect.
2. Key education stakeholders were not adequately sensitised on CBC.
3. Most teachers were unprepared to implement CBC.
4. Formative assessment tests were set by Kenya National Examinations Council. This is bad practice. Formative assessment tests are normally set and administered by class teachers

to obtain information for improvement of learners' performance.

Goal of CBC. The goal of CBC for primary schools is not appropriate, achievable, and realistic. Primary school children are too young to be equipped with competencies for employment. Preparing learners for employment is a goal of training institutions not schools. It is not the role of the school to prepare pupils for employment, but to make them trainable when they complete school. CBC is appropriate for technical training institutions not schools. Unemployment in Kenya is not caused by graduates lack of skills but by lack of employment opportunities. There are many doctors, teachers and technicians in Kenya who have no jobs.

Relevance of Subject Areas of CBC. Subject areas of CBC except foreign languages are relevant.

Extent to which CBC was Effectively Implemented in Primary Schools. The evidence from different sources indicates that:

1. Parental participation in their children's learning as proposed by CBC is too demanding and costly.
2. Teachers were not given adequate training on the subject matter of practical skills subjects.
3. Most people have not understood the importance of CBC.
4. Schools lack materials, equipment, tools and digital devices to effectively implement CBC because most parents are poor.
5. CBC makes heavy demands on learners and parents.
6. It is not realistic to expect parents to teach their children what they are not familiar with. This may lead to parents misleading their own children. Moreover many parents are semi-literate and cannot participate in their children's learning.
7. CBC does not put enough emphasis on the development of high level cognitive skills as is done in developed countries.
8. There is evidence that CBC is developing learners' ability to use ICT and digital devices and developing pupils' skills in the subjects they learn better than the previous curriculum.
9. Schools lack adequate equipment, tools and teaching/learning materials.
10. Some parents lack confidence in CBC.
11. Teachers' inability to implement CBC due to lack of adequate training.
12. Inadequate infrastructure and facilities.
13. The curriculum emphasises development of learners' competencies but teachers have not been adequately trained to develop practical skills and schools lack adequate equipment, tools and materials for practical skills. Hence CBC has not been effectively implemented. There are a number of reasons why schools lack adequate resources. Principal among these are financial constraints and inadequate funding by the government and high levels of poverty among parents who are expected to provide some learning materials.
14. CBC is not an appropriate curriculum framework for ba-

sis education.

15. There has been little systematic reform of the curriculum.

Unintended Outcomes of CBC. CBC has given rise to a number of unexpected outcomes. These include the following:

1. Many parents and education professionals have developed a negative attitude towards the school curriculum.

2. There is a belief among parents that their children will get employment when they complete school.

3. Contrary to the 1976 policy of using mother tongue as a medium of instruction in lower primary classes, currently English is used as a medium of instruction in all schools. Mother tongue is defined as the language of the catchment area.

Effects of CBC on Pupils. Due to implementation of CBC in primary schools, children have been equipped with practical skills.

Conclusions. On the basis of the findings of this study, the following conclusions are made:

1. The data suggests that the objectives of CBC are not realistic and achievable at school level.

2. The inadequate in-service training of teachers has hindered many of them from effectively implementing CBC.

3. It is clear from the results of this study that successful aspects of CBC are:

1. Use of Outcomes-based Curriculum Framework
2. Incorporation of continuous assessment scores in the final grading of pupils.
3. Use of learner-centred teaching methods.
4. Use of ICT.
5. Availability of adequate textbooks.

6. In general parents, head teachers, teachers and pupils have a positive attitude towards CBC.

Unsuccessful or less successful aspects of CBC include:

1. Not following the systematic process of curriculum reform.

2. Failure to pilot test curriculum for grades 4, 5 and 6.

3. Parental involvement of parents in their children's learning.

4. Resources for CBC implementation in most schools are inadequate for effective implementation of CBC.

5. Failure to carefully orientate Ministry of Education officials and parents to CBC.

6. Inadequate preparation of teachers prior to implementation of CBC.

7. Many learners from poor families are disadvantaged in ICT because their parents do not have smart phones and power connection.

8. Many teachers are not competent in handling practical skills subjects.

Recommendations

Based on the findings and conclusions of this study, the following recommendations are offered for improving CBC.

1. The government should ensure that CBC is evaluated and revised and a reasonable level of preparedness by schools and teachers to implement it is achieved in all parts of the country. However, there is no need to call it CBC.

2. The government should post fully trained technical teachers to primary schools if it wants to develop pupils' competencies.

3. Involvement of parents in their children's learning should be reduced.

4. The Ministry of Education should work closely with teacher training colleges in conducting in-service teacher training workshops.

5. The Ministry of Education should focus its resources on one or more of the following areas:

1. Employing more teachers.
2. Developing competence-based profile of teachers in conjunction with teacher training colleges.
3. Providing equipment, tools and coming up with materials to schools.

Replication of the Study. Kenya Institute of Curriculum Development should commission a consultancy firm or team of professional evaluators to replicate this study in more counties.

Recommendation for the Future Evaluation. A large scale independent formative evaluation of implementation of CBC in junior secondary schools should be conducted.

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